

COURSE MATERIAL

SUBJECT	ENGINEERING GRAPHICS (ME23AES102)
UNIT	3
COURSE	B.TECH
SEMESTER	11
DEPARTMENT	MECHANICAL ENGINEERING
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TABLE OF CONTENTS – UNIT 3

SNO	CONTENTS	PAGE
1	COURSE OBJECTIVES	3
2	PREREQUISITES	3
3	SYLLABUS	3
4	COURSE OUTCOMES	3
5	CO - PO/PSO MAPPING	3
6	LESSON PLAN	4
7	LECTURE NOTES	5
	3.1 INTRODUCTION	3
	3.2 PROJECTION OF SOLIDS	13
8	ASSIGNMENTS	26
9	PART B QUESTIONS	27
10	SUPPORTIVE ONLINE CERTIFICATION COURSES	27
11	CONTENTS BEYOND THE SYLLABUS	27
12	PRESCRIBED TEXT BOOKS & REFERENCE BOOKS	27

1. COURSE OBJECTIVES

The objectives of this course is to introduce and make the students

1. To enable with various concepts like dimensioning, conventions and standards related to Engineering Drawing
2. To impart knowledge on the Projection of Points, Lines and Plane surfaces
3. To improve the visualization skills for better understanding of projection of solids
4. To develop the imaginative skills required to understand Section of solids and Developments of surfaces.
5. To understand the viewing perception of a solid object in Isometric and orthographic projections.

6. PREREQUISITES

Students should have knowledge on
Basic Mathematics

7. SYLLABUS**UNIT III**

Projections of Solids: Types of Solids: Polyhedra and Solids of Revolution. Projections of Solids in Simple Positions: Axis Perpendicular to Horizontal Plane, Axis Perpendicular to Vertical Plane and Axis Parallel to both the Reference Planes, Projection of Solids with Axis Inclined to One Reference Plane and Parallel to another Plane, Projection of Solids Inclined to both the Reference Planes..

8. COURSE OUTCOMES

After completing the course, the student will be able to

1. Draw various engineering curves, scales. (L3)
2. Draw and Interpret orthographic projections of points, lines, planes. (L3)
3. Draw the projection of solids in various positions. (L3)
4. Draw and Explore the sections of solids and development of surfaces. (L3)
5. Draw an isometric and orthographic views of simple solids. (L3)

9. Co-PO / PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	2	2								2		2	2	2
CO2	2	2								2		2	2	2
CO3	2	2								2		2	2	2
CO4	2	2								2		2	2	2
CO5	2	2			1					2		2	2	2

10. LESSON PLAN

LECTURE	WEEK	TOPICS TO BE COVERED	REFERENCES
1	1	Introduction to solids	T1
2		Difference between prism and pyramids	T1, R1
3		Projection of solids	T1, R1
4		Projection of solids by rotational method	T1, R2
5	2	Projection of solids inclined to both the planes	T1, R1
6		Practice	T2, R1
7		Practice	T2, R1



11. LECTURE NOTES

3.1 INTRODUCTION

A plane is a two-dimensional object having length and breadth only. Its thickness is always neglected; various shapes of plane figures are considered such as square, rectangle, circle, pentagon, hexagon, etc.

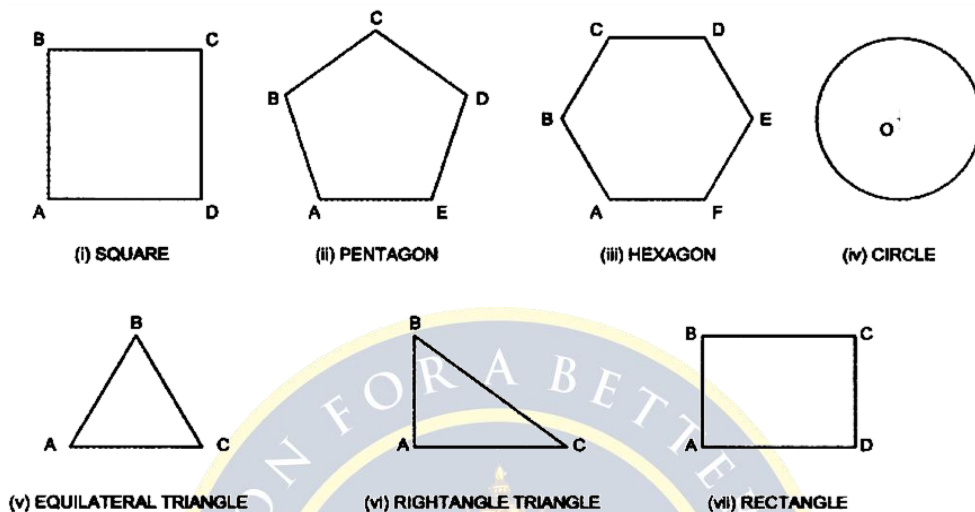


Figure 3.1

3.1.1 TYPES OF PLANES:

- Perpendicular planes which have their surface perpendicular to anyone of the reference planes and parallel or inclined to the other reference plane.
- Oblique planes which have their surface inclined to both the reference planes.

3.1.2 TRACE OF PLANE:

The trace of a plane is the line of intersection or meeting of the plane surface with the reference plane, if necessary, the plane surface is extended to intersect the reference plane. The intersection line of the plane surface with HP is called the Horizontal Trace (HT) and that of VP is called the Vertical Trace (VT).

A plane figure is positioned with reference to the reference planes by referring its surface in the following possible position.

Projection of Different Planes position with respect to Principal planes

- 1) Surface of Plane Parallel to the HP (and perpendicular to VP)

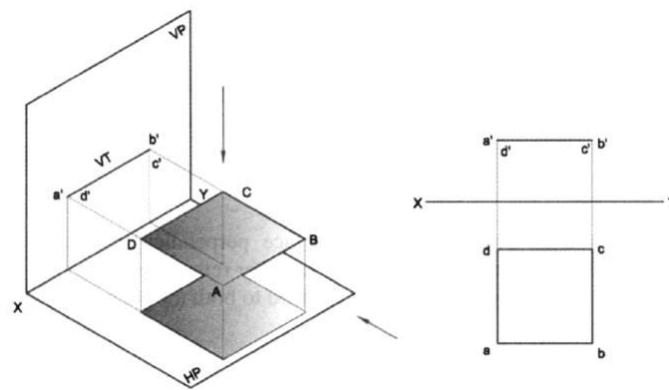


Figure 3.2

2) Surface of Plane Parallel to the VP (and perpendicular to HP)

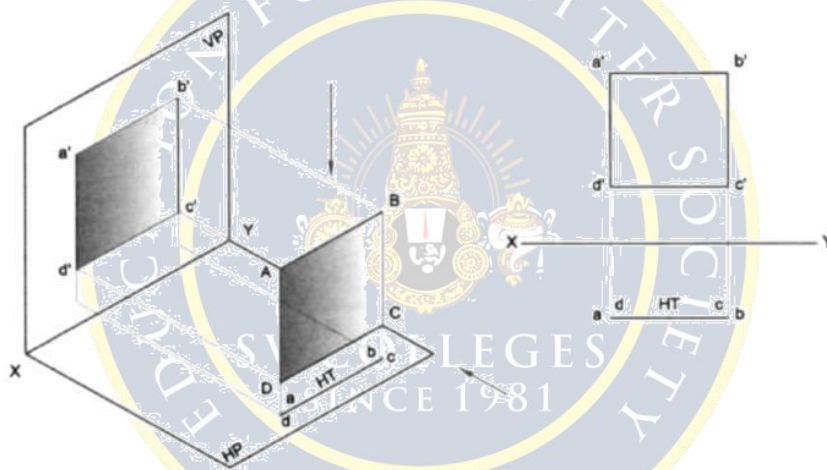


Figure 3.3

3) Surface of Plane Inclined to the HP and perpendicular to VP

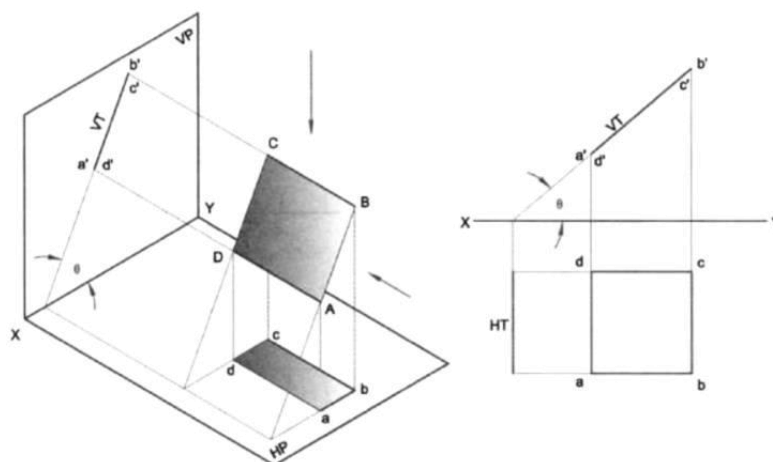


Figure 3.4

4) Surface of Plane Inclined to the VP and perpendicular to HP

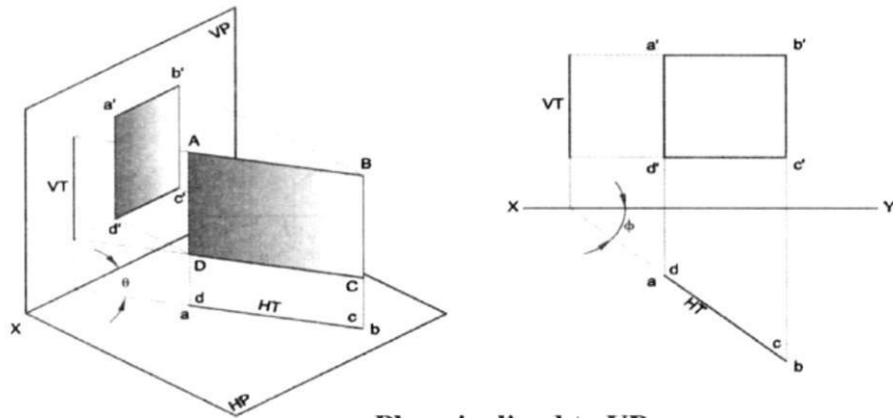


Figure 3.5

5) Surface of Plane Perpendicular to Both HP and VP

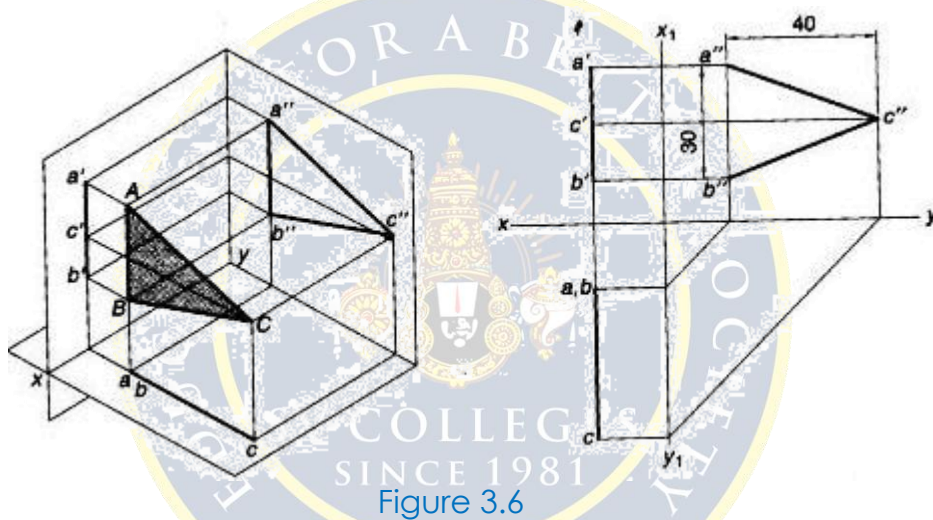


Figure 3.6

6) Surface of Plane is Inclined to Both HP and VP

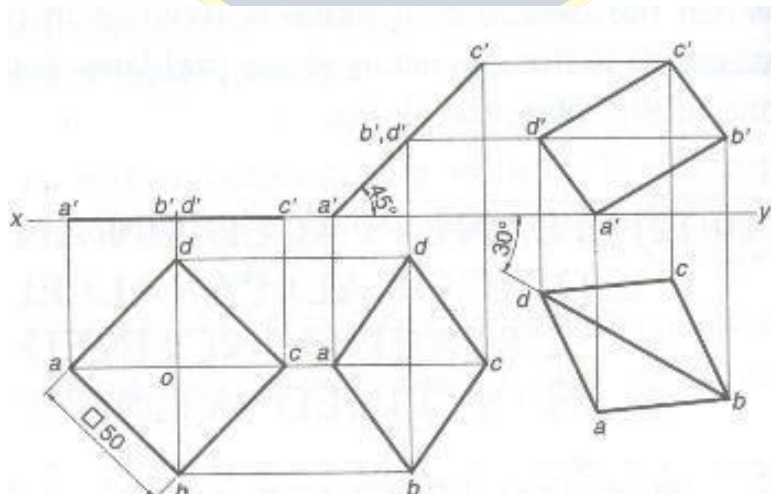


Figure 3.7

3.1.3 Simple Problems:

Plane Parallel to HP

Problem: A Square plane with a 40mm side has its surface parallel to and 20mm above the HP. Draw its Projections, when (a) A side is parallel to VP (b) A side is inclined at 30° to VP and (c) All sides are equally inclined to VP.

Visualized position of surface plane Picture:

Solution:

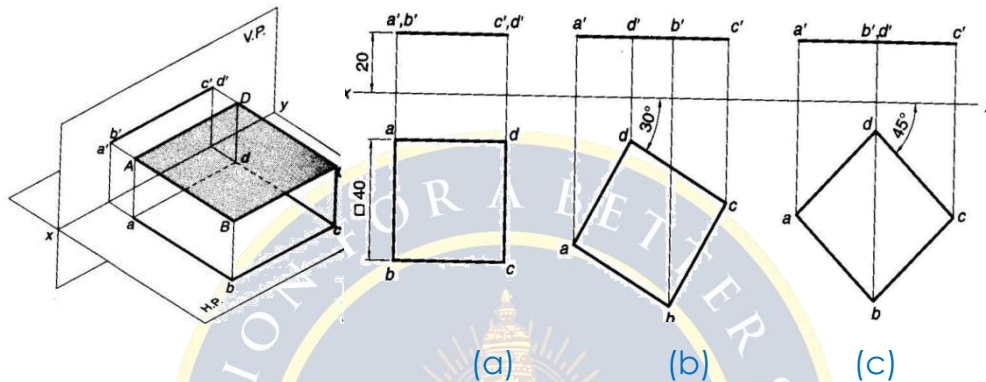


Figure 3.8(a)

Figure 3.8(b)

Plane Parallel to Plane Parallel to VP

Problem: A Hexagonal plane with a 30mm side has its surface parallel to and 20mm in front of the VP. Draw its Projections, when (a) a side is perpendicular to HP (b) a side is parallel to the HP (c) Side is inclined at 45° to the HP.

Visualized position of surface plane Picture:

Solution:

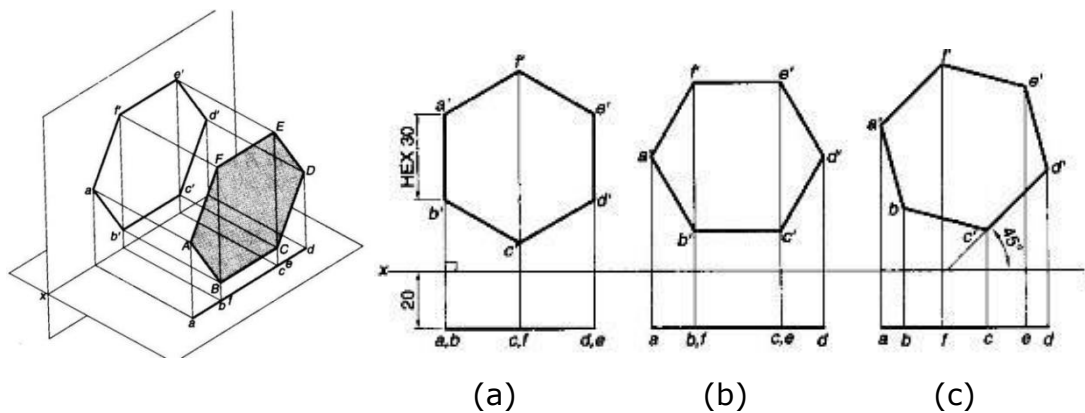


Figure 3.9 (a)

Figure 3.9(b)

Plane is inclined to HP and Perpendicular to VP

Problem: A Pentagonal plane with a 30mm side has an edge on the HP, the surface of the Plane is inclined at 45° to the HP. Draw its Projections?

Visualized position of surface plane Picture:

Solution:

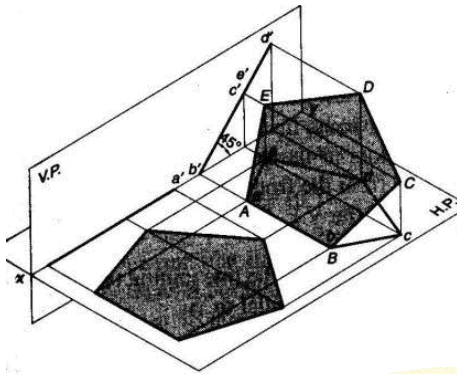


Figure 3.10(a)

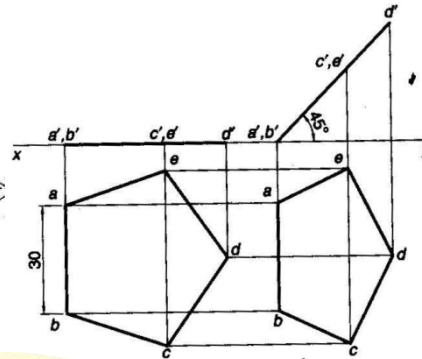


Figure 3.10(b)

Problem: A Hexagonal plate with a 30mm side and negligible thickness has its surface perpendicular to the HP and inclined at 45° to the VP. Draw its Projections? When one of its sides of the Plane is Parallel to and 15 mm in front of the VP.

Visualized position of surface plane Picture:

Solution:

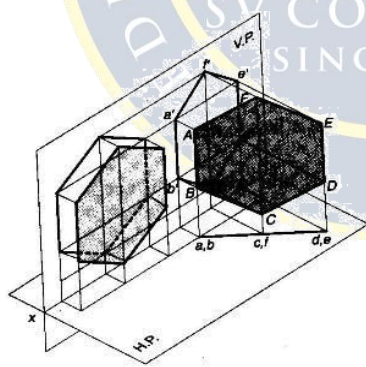


Figure 3.11(a)

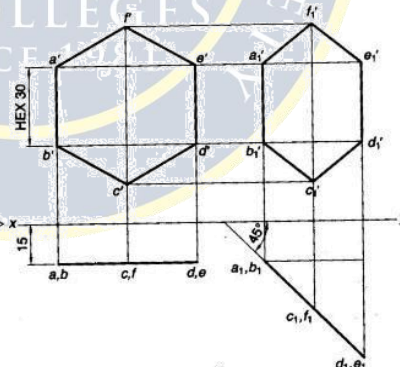


Figure 3.11(b)

Problem: A Circular plane with a 60mm Diameter is resting on a point its circumference on the VP. The centre is 40 mm above the HP, and the surface is inclined at 45° to the VP. And perpendicular to the HP Draw Its Projections?

Solution:

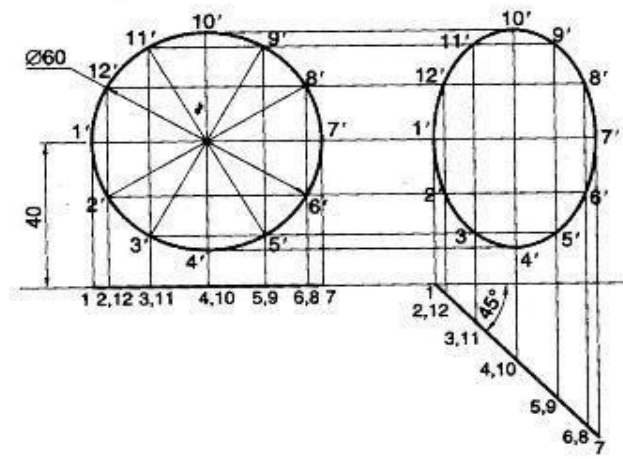


Figure 3.12

Problem: Rectangle 30mm and 50mm sides is resting on HP on one of its small side, which is 300 inclined to VP, while the surface of the plane makes 450 inclination with HP. Draw it is projections?

Solution:

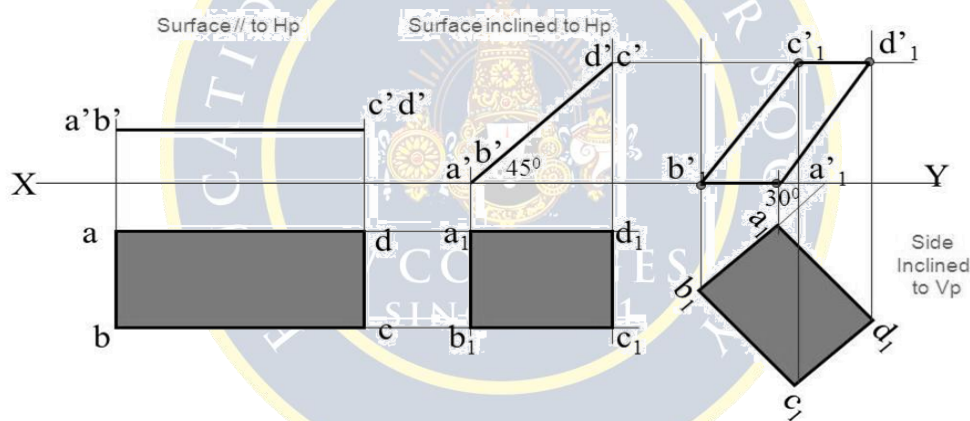


Figure 3.13

Problem: A regular pentagon of 30 mm sides is resting on HP, on one of its sides with its surface 450 inclined to HP. Draw it is projections when the side in HP makes 300 angle with VP?

Solution:

According to the given Problem

1. Surface inclined to HP plane
2. Assumption for initial position is parallel to HP.
3. So, TV view will show True shape. Hence begin with TV, draw pentagon below X-Y line, taking one side vertical.

Note: Surface and side inclination are directly given

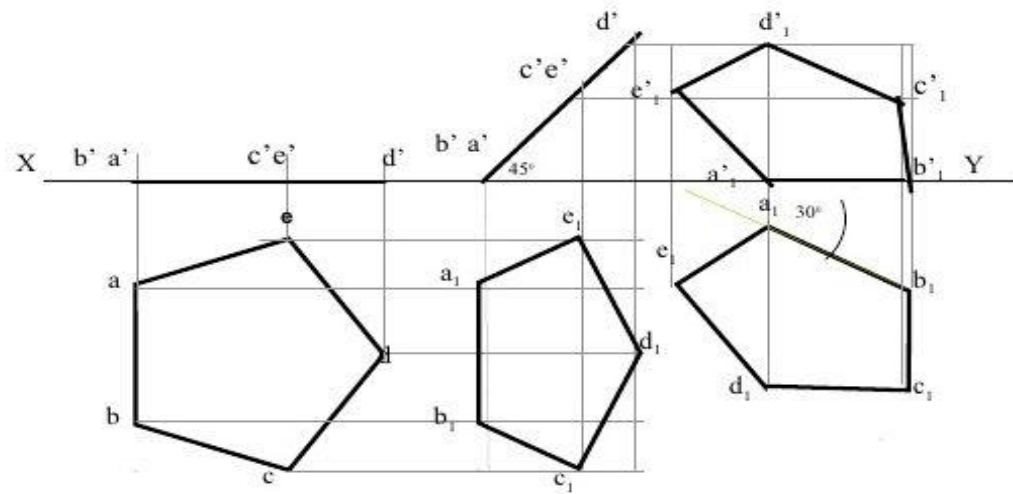


Figure 3.14

Problem: A regular pentagon of 30 mm sides is resting on HP on one of its sides while it is opposite vertex (corner) is 30 mm above HP. Draw projections when side in HP is 30° inclined to VP.

Solution:

According to the given Problem

Surface inclined to HP plane.

Assumption for initial position is parallel to HP.

So, TV view will show True shape. Hence begin with TV, draw pentagon below X-Y line, taking one side vertical.

Note: Surface Inclination indirectly given, and side inclination is Directly given only change is the way surface inclination is described: One side on HP & it is opposite corner 30 mm above HP. Hence redraw 1st Fv as a 2nd Fv making above arrangement. Keep $a'b'$ on xy & d' 30 mm above xy.

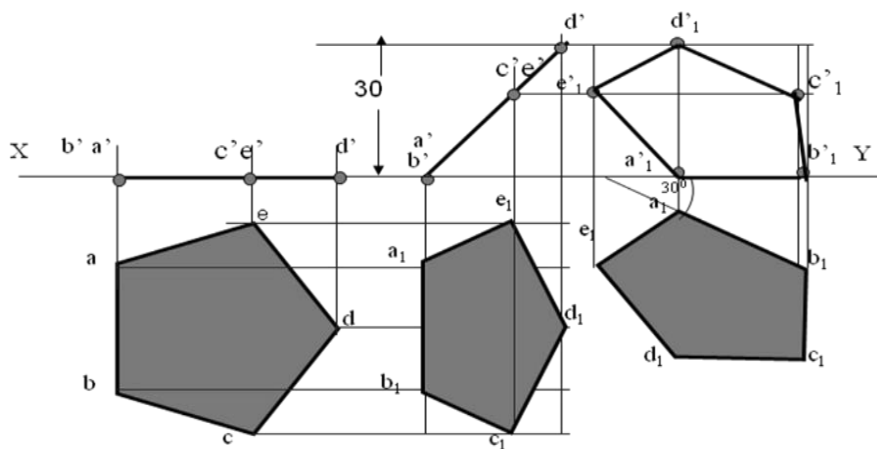


Figure 3.15

Problem: A circle of 50 mm diameter is resting on HP on end A of its diameter AC which is 30° inclined to HP while its TV is 45° inclined to VP. Draw its Projections?

Solution:

According to the given Problem

Surface inclined to HP plane.

Assumption for initial position parallel to HP.

So which TV will show True shape?

Which diameter AC horizontal Hence begin with TV, draw rhombus below X-Y line, taking longer diagonal parallel to X-Y.

Note: In This problem inclination of TV of that AC is given, It could be drawn directly as shown in 3rd step. of that AC is given, It could be drawn directly as shown in 3rd step.

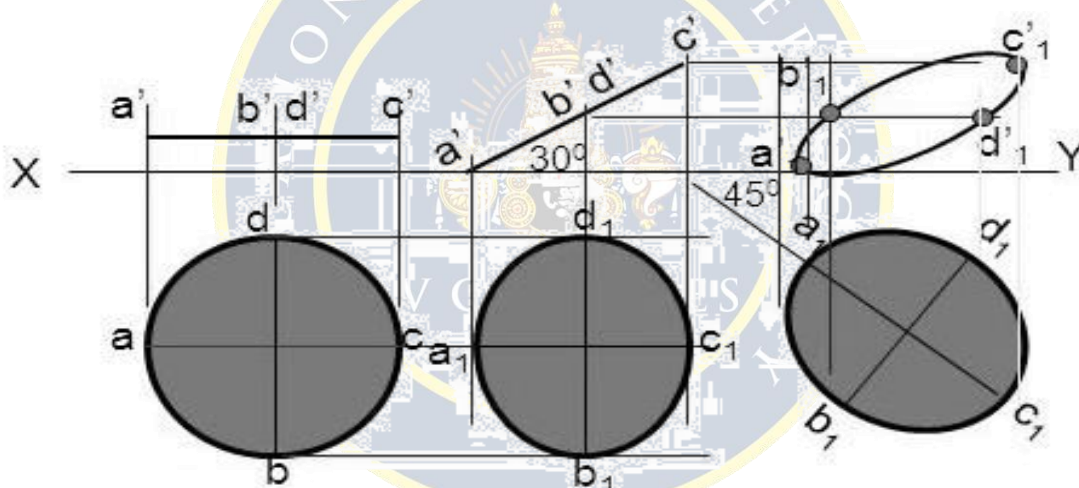


Figure 3.16

3.1.4 Important Questions:

1. A pentagon of sides 30mm rests on the ground on one of its corners with the sides containing the corners being equally inclined to the ground. The side opposite to the corner on which it rests is inclined at 30 degrees to the VP and is parallel to the HP. The surface of the pentagon makes 10 degrees with the ground. Draw the top and front views of the pentagon.
2. A regular pentagon of 30mm side is resting on one of its edges on HP which is inclined at 45 degrees to VP. Its surface is inclined at 30 degrees to HP. Draw its projections.

Previous Paper Questions:

1. Draw the projections of a regular hexagon of 25mm side, having one of its sides in the H.P. and inclined at 60 degrees to the V.P., and its surface making an angle of 45 degrees with H.P.
2. A thin circular plate of 40mm diameter having its plane vertical and inclined at 40 Its centre is 30mm above H.P. and 35mm in front of V.P. Draw the projections to V.P.

3.2 PROJECTION OF SOLIDS**3.2.1 Introduction:**

A solid has three dimensions, the length, breadth and thickness or height. A solid may be represented by orthographic views, the number of which depends on the type of solid and its orientation with respect to the planes of projection. Solids are classified into two major groups. (i) Polyhedral, and (ii) Solids of revolution

Polyhedral: A polyhedral is defined as a solid bounded by plane surfaces called faces. They are: (i) Regular polyhedral (ii) Prisms and (iii) Pyramids

Regular Polyhedral : A polyhedron is said to be regular if its surfaces are regular polygons. The following are some of the regular polyhedral.

SOLIDS

Prisms: A prism is a polyhedron having two equal ends called the bases parallel to each other. The two bases are joined by faces, which are rectangular in shape. The imaginary line passing through the centres of the bases is called the axis of the prism.

A prism is named after the shape of its base. For example, a prism with square base is called a square prism, the one with a pentagonal base is called a pentagonal prism, and so on (Fig) The nomenclature of the prism is given in Fig.

To understand and remember various solids in this subject properly, those are classified & arranged in to two major groups.

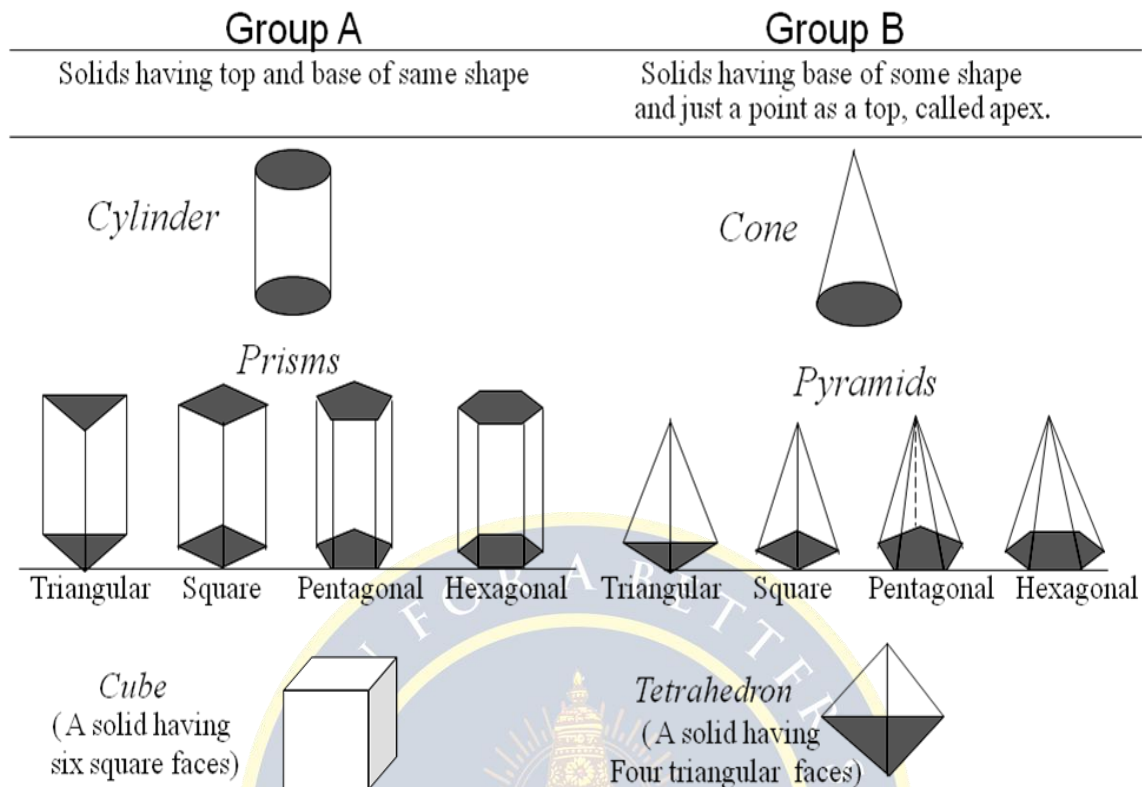


Figure 3.18

- Tetrahedron:** It consists of four equal faces, each one being an equilateral triangle.
- Hexa Hedron (Cube):** It consists of six equal faces, each a square.
- Octahedron:** It has eight equal faces, each an equilateral triangle.
- Dodecahedron:** It has twelve regular and equal pentagonal faces.
- Icosahedrons:** It has twenty equal, equilateral triangular faces.

Pyramids: A pyramid is a polyhedron having one base, with several isosceles triangular faces, meeting at a point called the apex. The imaginary line passing through the centre of the base and the apex is called the axis of the pyramid.

The pyramid is named after the shape of the base. Thus, a square pyramid has a square base and pentagonal pyramid has pentagonal base and so on. The nomenclature of a pyramid is shown in Fig.

Dimensional parameters of different solids.

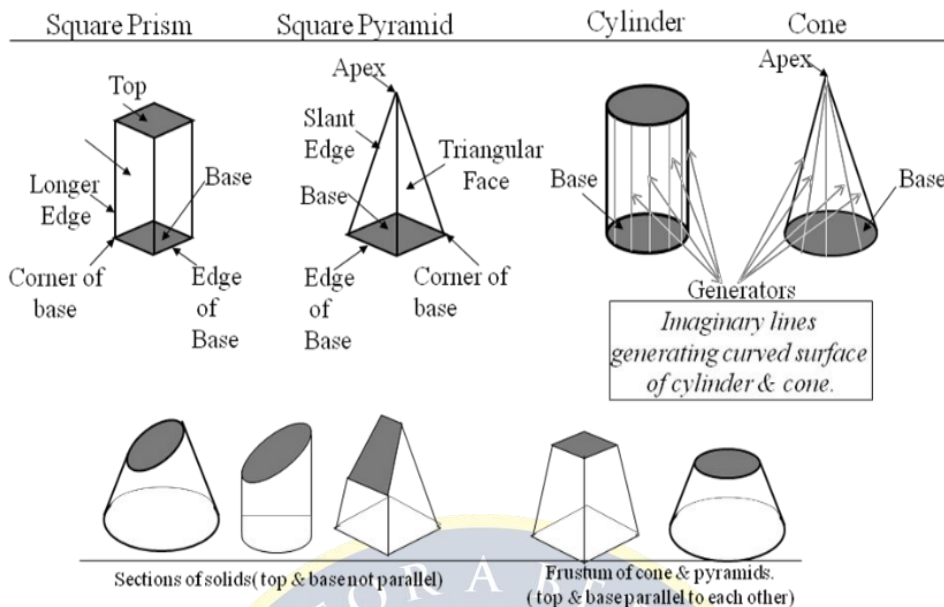


Figure 3.19

3.2.2 Types of Pyramids:

There are many types of Pyramids, and they are named after the shape of their base. These are Triangular Pyramid, Square Pyramid, Pentagonal pyramid, hexagonal pyramid and tetrahedron

Solids of Revolution: If a plane surface is revolved about one of its edges, the solid generated is called a solid of revolution. The examples are (i) Cylinder, (ii) Cone, (iii) Sphere.

Frustums and Truncated Solids: If a cone or pyramid is cut by a section plane parallel to its base and the portion containing the apex or vertex is removed, the remaining portion is called frustum of a cone or pyramid.

Prisms Position of a Solid with Respect to the Reference Planes: The position of solid in space may be specified by the location of either the axis, base, edge, diagonal, or face with the principal planes of projection. The following are the positions of a solid considered.

Axis perpendicular to HP

Axis perpendicular to VP

Axis parallel to both the HP and VP

Axis inclined to HP and parallel to VP

Axis inclined to VP and parallel to HP

Axis inclined to both the Planes (VP. and HP)

The position of solid with reference to the principal planes may also be grouped as follows:

Solid resting on its base.

Solid resting on anyone of its faces, edges of faces, edges of base, generators, slant edges, etc.

Solid suspended freely from one of its corners, etc.

1. Axis perpendicular to one of the principal planes:

When the axis of a solid is perpendicular to one of the planes, it is parallel to the other. Also, the projection of the solid on that plane will show the true shape of the base. When the axis of a solid is perpendicular to H.P, the top view must be drawn first and then the front view is projected from it. Similarly, when the axis of the solid is perpendicular to V.P, the front view must be drawn first and then the top view is projected from it.

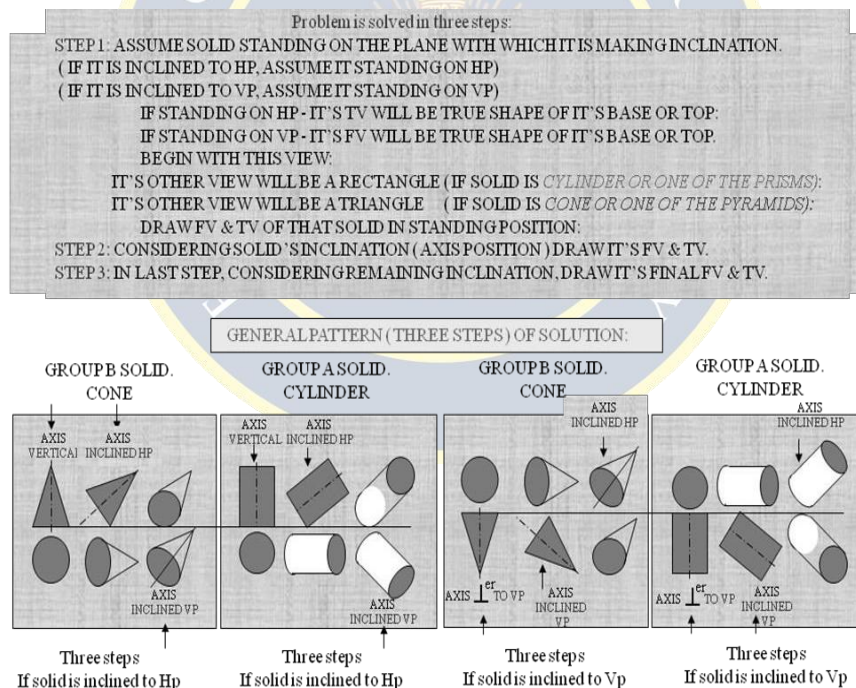


Figure 3.20

3.2.3 Simple Problems:

When the axis of solid is perpendicular to one of the planes, it is parallel to the other. Also, the projection of the solid on that plane will show the true shape of the base. When the axis of a solid is perpendicular to H.P, the top view must be drawn first and then the front view is projected from it. Similarly, when the

axis of the solid is perpendicular to V.P, the front view must be drawn first and then the top view is projected from it.

1. Axis perpendicular to HP

Problem: A Square Pyramid, having base with a 40 mm side and 60mm axis is resting on its base on the HP. Draw its Projections when (a) a side of the base is parallel to the VP. (b) A side of the base is inclined at 30° to the VP and (c) All the sides of base are equally inclined to the VP.

Solution:

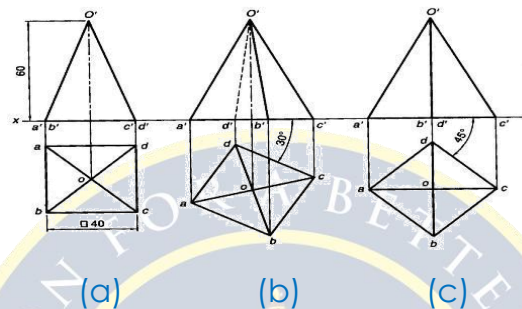


Figure 3.21

2. Axis perpendicular to VP

Problem: A pentagonal Prism having a base with 30 mm side and 60mm long Axis, has one of its bases in the VP. Draw Its projections When (a) rectangular face is parallel to and 15 mm above the HP (b) A rectangular face perpendicular to HP and (c) a rectangular face is inclined at 45° to the HP

Solution:

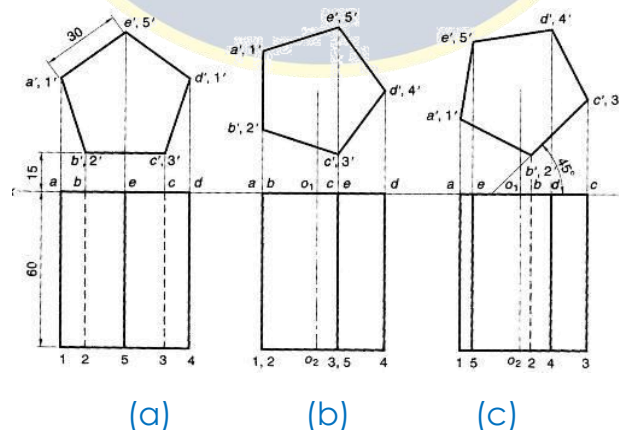


Figure 3.22

3. Axis parallel to both the HP and VP

Problem: A pentagonal Prism having a base with a 30 mm side and 60mm long axis, is resting on one of its rectangular faces on the HP. with axis parallel

to the VP. Draw its projections?

Solution:

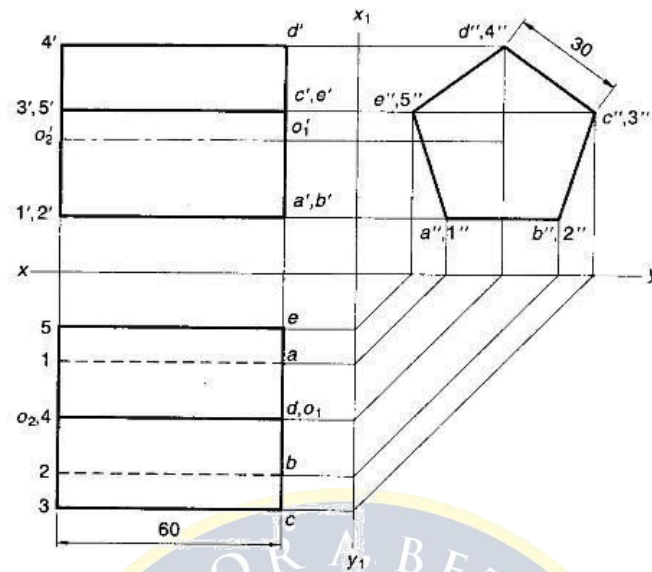


Figure 3.23

4. Axis inclined to HP and parallel to VP

Problem: A Hexagonal Prism having a base with a 30 mm side and 75 mm long axis, has an edge its base on the HP. Its axis is Parallel to the VP and inclined at 45° to the HP. Draw its projections?

Solution:

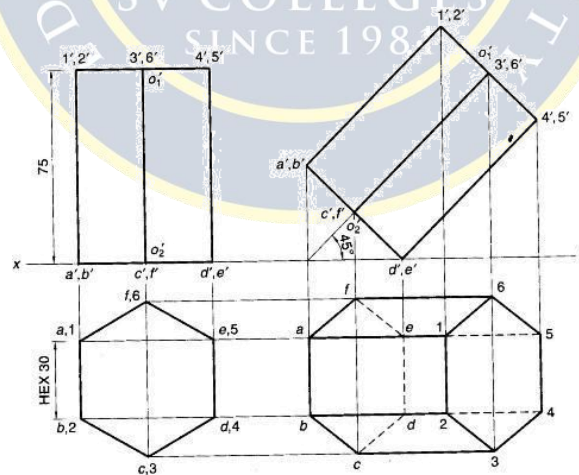


Figure 3.24

5. Axis inclined to VP and parallel to HP

Problem: A Hexagonal Prism, having a base with a 30 mm side and 65 mm long axis, has an edge its base in the VP Such that the axis is inclined at 30° to the VP and Parallel to the HP. Draw its Projections?

Solution:

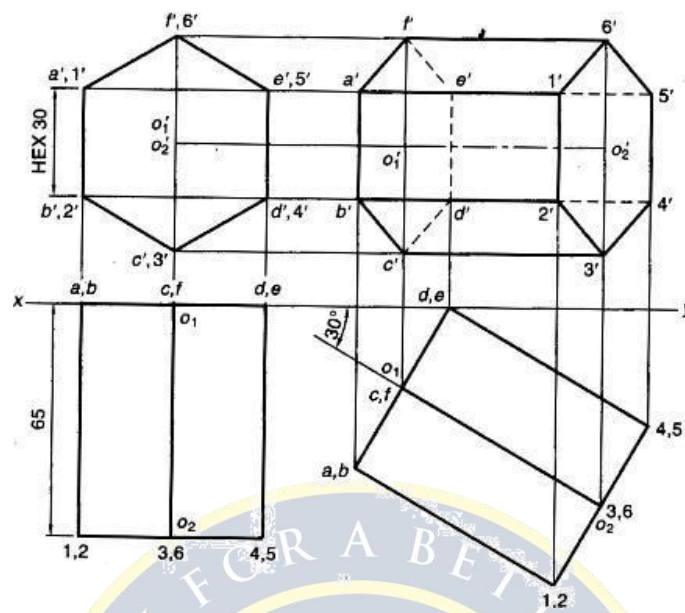


Figure 3.25

6. Axis inclined to both the principal planes (HP and VP)

A solid is said to be inclined to both the planes when (i) the axis is inclined to both the planes, (ii) the axis is inclined to one plane and an edge of the base is inclined to the other. In this case the projections are obtained in three stages.

Stage I: Assume that the axis is perpendicular to one of the planes and draw the projections.

Stage II: Rotate one of the projections till the axis is inclined at the given angle and project the other view from it.

Stage III: Rotate one of the projections obtained in Stage II, satisfying the remaining condition and project the other view from it.

Problem: A cube of 50 mm long edges is so placed on HP on one corner that a body diagonal is Parallel to HP and perpendicular to VP. Draw its projections.

Solution Steps:

Assuming standing on HP, begin with TV, a square with all sides equally inclined to xy. Project FV and name all points of FV & TV.

Draw a body-diagonal joining c' with 3' (This can become Parallel to xy)

1. From $1'$ drop a perpendicular on this and name its p' .
2. Draw 2nd FV in which $1'-p'$ line is vertical means $c'-3'$ diagonal must be horizontal. Now as usual project TV.
3. In final TV draw same diagonal is perpendicular to VP as said in problem. Then as usual project final FV.

Solution:

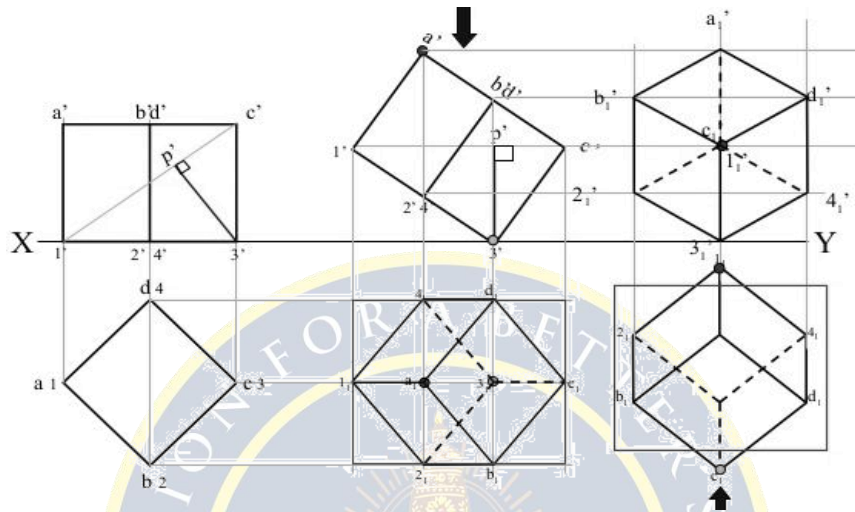


Figure 3.26

Problem: A cone 40 mm diameter and 50 mm axis is resting on one of its generator on HP which makes 30° inclinations with VP. Draw its projections?

Solution Steps:

Resting on HP on one generator, means lying on HP

1. Assume it standing on HP.
2. Its TV will show True Shape of base (Circle)
3. Draw 40mm dia. Circle as TV & taking 50 mm axis project FV. (a triangle)
4. Name all points as shown in illustration.
5. Draw 2nd FV in lying position i.e., $o'e'$ on xy . And project it is TV below xy .
6. Make visible lines dark and hidden dotted, as per the procedure.
7. Then construct remaining inclination with VP (Generator $o1e1$ 30° to xy as shown) & project final FV.

Solution:

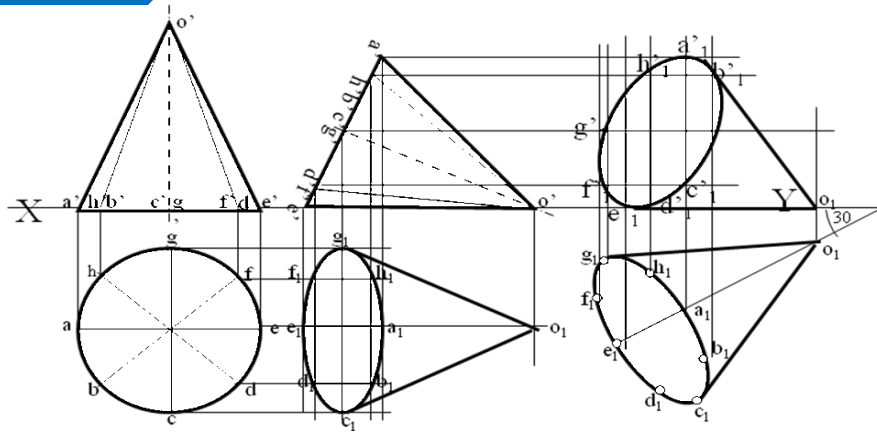


Figure 3.27

Problem: A cube of 50 mm long edges is so placed on HP on one corner that a body diagonal through this corner is perpendicular to HP and parallel to VP. Draw its three views.

Solution Steps:

1. Assuming it is standing on HP begin with TV, a square of corner case.
2. Project corresponding FV & name all points as usual in both views.
3. Join $a'l'$ as body diagonal and draw 2nd FV making it vertical (l' on xy)
4. Project its TV drawing dark and dotted lines as per the procedure.
5. With standard method construct Left-hand side view. (Draw a 45° inclined line in Tv region (below xy). Project horizontally all points of Tv on this line and reflect vertically upward, above xy . After this, draw horizontal lines, from all points of Fv, to meet these lines. Name points of intersections and join properly. For dark & dotted lines locate observer on left side of Fv as shown.)

Solution:

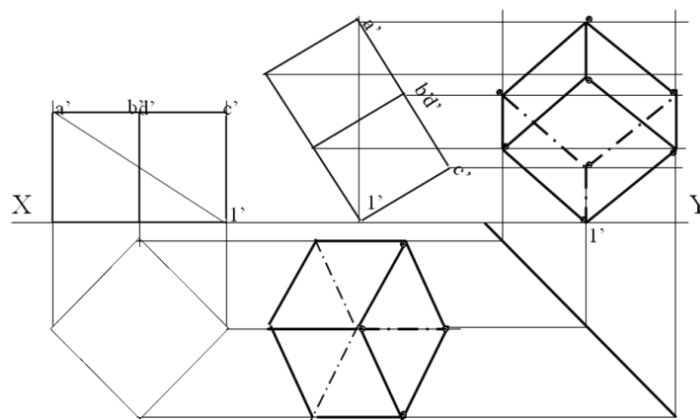


Figure 3.28

Problem: A circular cone, 40 mm base diameter and 60 mm long axis is resting on HP, on one point of base circle such that its axis makes 45° inclination with HP and 40° inclination with VP. Draw its projections.

Solution:

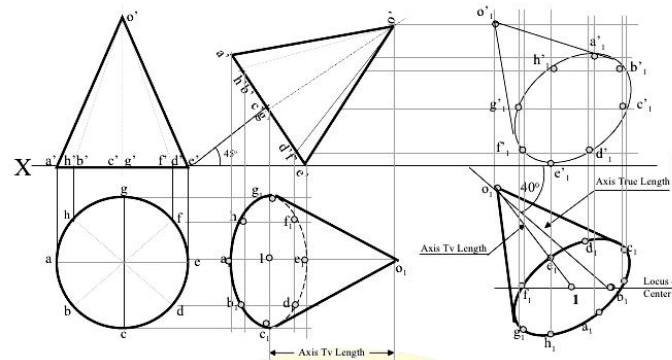


Figure 3.29

Problem: A hexagonal prism, having a base with a 30mm side and an 80mm long axis, rests on one of its base edges in the H.P such that the axis is inclined at 30° to the HP and 45° to the VP. Draw its projections?

Solution:

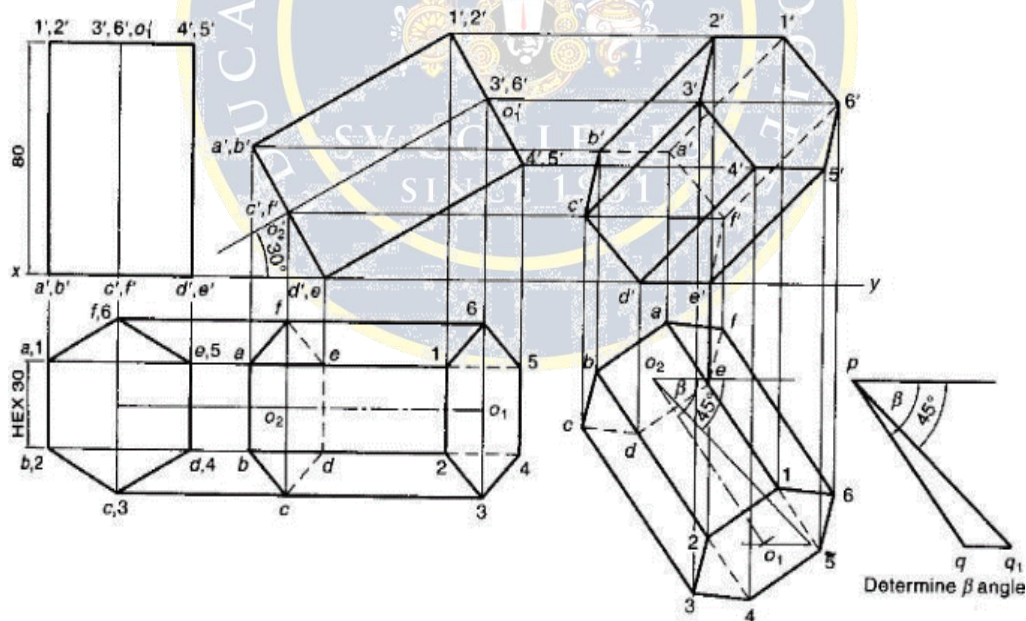


Figure 3.30

Problem: A Square prism, having a base with a 35mm side and an 60mm long axis, rests on one of its base edges in the HP such that the axis is inclined at 45° to the HP and 45° to the VP. Draw its projections if the resting edge makes an angle of 30° with VP?

Solution:

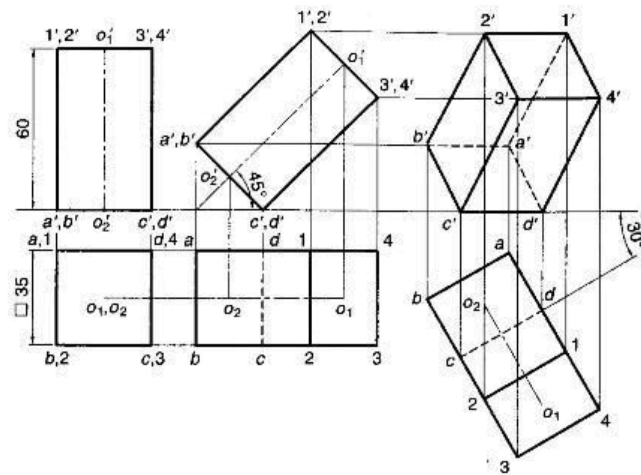


Figure 3.31

Problem: A cylinder 40 mm diameter and 50 mm axis is resting on one point of a base circle on VP while its axis makes 45° with VP and FV of the axis 35 with HP. Draw its projections.

Solution Steps:

Resting on VP on one point of base, means inclined to VP:

1. Assume it standing on VP
2. Its FV will show True Shape of base & top (circle)
3. Draw 40mm dia. Circle as FV & taking 50 mm axis project TV.
(a Rectangle)
4. Name all points as shown in illustration.
5. Draw 2nd TV making axis 45° to xy and project it as FV above xy.
6. Make visible lines dark and hidden dotted, as per the procedure.
7. Then construct remaining inclination with HP (FV of axis i.e., centre line of view to xy as shown) & project final.

Solution:

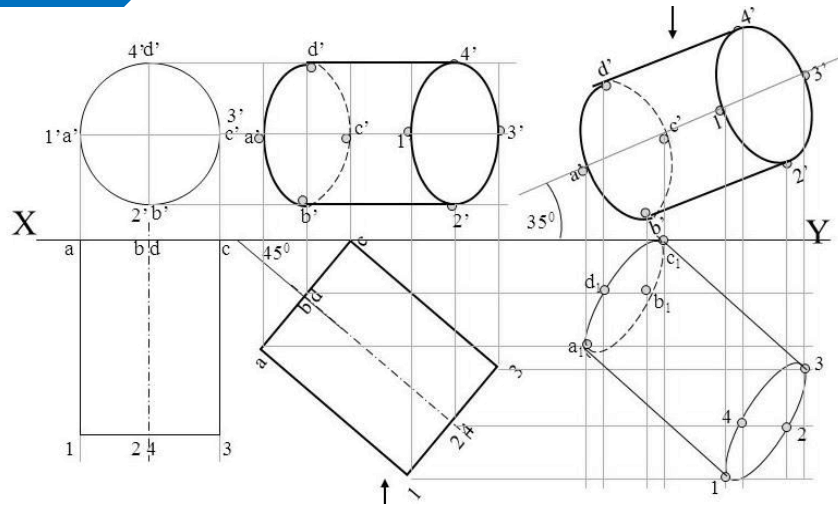


Figure 3.32

Problem: A hexagonal prism of base side 30 mm long and axis 40 mm long, is standing on HP on its base with one base edge parallel to VP. A tetrahedron is placed centrally on the top of it. The base of tetrahedron is a triangle formed by joining alternate corners of top of prism. Draw projections of both solids. Project an auxiliary TV on AIP 45° inclined to HP.

Solution:

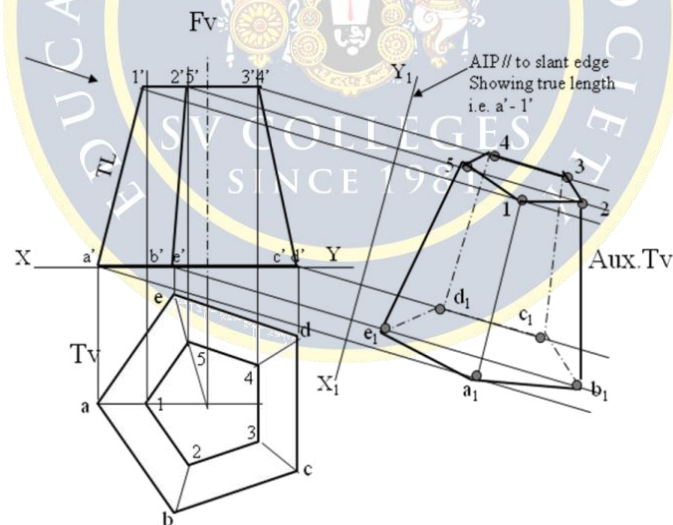


Figure 3.33

Tips & Shortcuts:

Axis inclined to HP and Parallel to $\rightarrow$ VP must solve in two stages

Stage(i) assume axis perpendicular to HP then draw Top and Front view

Stage(ii) Tilt the Front view according to given angle. Then project all the points will get Final Top view

Axis inclined to VP and Parallel to HP $\rightarrow$ must solve in two stages

Stage(i) assume axis perpendicular to VP then draw front and Top view

Stage(ii) Tilt the Top view according to given angle. Then project all the points will get Final Front view

Previous paper questions and Important Problems:

A cone of base diameter 40 mm and axis height 60 mm rests on the ground on a point of its base circle such that the axis of the cone is inclined at 40° to the HP and 30° to the VP. Draw its front and top views.

A hexagonal prism of base of side 40 mm and axis length 80 mm rests on one of its base edges on the HP. The end containing that edge is inclined at 30° to the HP and the axis is parallel to VP. It is cut by a plane perpendicular to the VP and parallel to the HP. The cutting plane bisects the axis. Draw its front and the sectional top views.

A square pyramid of base side 30 mm and altitude 50 mm lies on one of its triangular faces on the HP with its axis parallel to the VP. It is cut by a vertical plane inclined at 30° to the VP and meeting the axis at 40 mm from the vertex measured in the plan. Draw the top view, sectional front view and the true shape of the section.

A cone, diameter of base 50 mm and axis 65 mm long, is lying on the HP, on one of its generators with the axis parallel to the VP. It is cut by a horizontal Section plane 12mm above the ground. Draw its front view and sectional top view.

Draw the projections of a hexagonal pyramid of side of base 30mm and axis 60mm long resting on one of its base edges in HP with its axis inclined at 30° to HP, and the top view of axis is 45° to VP.

A square prism having a base with a 40mm side and a 60 mm long axis rests on its base on HP, such that one of the vertical faces makes an angle of 30° degrees with VP. A section plane perpendicular to the VP, inclined at 45° degrees to the HP, and passing through the axis at a point 20 mm from its top end cuts the prism. Draw its front view, sectional top view.

A hexagonal prism, side of base 35mm and height 75mm is resting on one of its corners on HP, with a longer edge containing that corner inclined at 60° degrees to the HP, and rectangular face parallel to the VP. A horizontal

section plane cuts the prism into two halves. Draw the sectional top view of the cut prism and front view.

Draw the projections of a pentagonal prism, base 25 mm side and axis 50 mm long resting on one of its rectangular faces on HP, with the axis inclined at 45 degrees to VP.

12. Assignments

S.No	Question	BL	CO
1	A square prism, side of base 30mm and axis 45mm long lies on H.P. such that its axis is parallel to both H.P. and V.P. Draw the top and front views of the prism when i It lies with one of its rectangular faces on H.P. and ii It lies with one of its longer edges on H.P.	1	3
2	iii Draw the projections of a cylinder of base 30mm diameter and axis 50mm long when it is resting on HP on one of its bases. iv A cube of 40mm side is resting with a face on HP such that, the vertical faces are equally inclined to VP. Draw its projections	1	3
3	A pentagonal prism is resting on one of the corners of its base on HP. The longer edge containing that corner is inclined at 30° to HP and the vertical plane containing that edge is inclined at 45° to VP. Draw the projections of the solid. Take the side of the base as 30 and length of the axis as 80mm.	1	3
4	Draw the projections of a cone of 100mm height and 75 diameters, resting on one of its generators on HP and its axis lies in a vertical plane, inclined at 30° to VP.	1	3
5	A Hexagonal Pyramid side of base 30mm and axis 50mm long rests with one of the corners of its base on H.P. Its axis is inclined at 35° to H.P. and 45° to V.P. Draw its projections.	1	3

13. Part B- Questions

S.No	Question	BL	CO
1	A square prism, side of base 30mm and axis 45mm long lies on H.P. such that its axis is parallel to both H.P. and V.P. Draw the top and front views of the prism when It lies with one of its rectangular faces on H.P. and It lies with one of its longer edges on H.P.	1	3
2	i. Draw the projections of a cylinder of base 30mm diameter and axis 50mm long when it is resting on HP on one of its bases. ii. A cube of 40mm side is resting with a face on HP such that, the vertical faces are equally inclined to VP. Draw its projections	1	3
3	A pentagonal prism is resting on one of the corners of its base on HP. The longer edge containing that corner is inclined at 30° to HP and the vertical plane containing that edge is inclined at 45° to VP. Draw the projections of the solid. Take the side of the base as 30 and length of the axis as 80mm.	1	3
4	Draw the projections of a cone of 100mm height and 75 diameters, resting on one of its generators on HP and its axis lies in a vertical plane, inclined at 30° to VP.	1	3
5	A Hexagonal Pyramid side of base 30mm and axis 50mm long rests with one of the corners of its base on H.P. Its axis is inclined at 35° to H.P. and 45° to V.P. Draw its projections.	1	3

14. Supportive Online Certification Courses

1. Engineering drawing and computer graphics By Prof. Rajaram Lakkaraju, conducted by IIT Kharagpur – 12 weeks

15. Prescribed Textbooks & Reference Books

Textbook

1. K.L. Narayana & P. Kanniah, Engineering Drawing, 3/e, SciTech Publishers, Chennai, 2012.
2. N.D. Bhatt, Engineering Drawing, 53/e, Charotar Publishers, 2016.

16. References:

1. Venugopal, Engineering Drawing and Graphics, 3/e, New Age Publishers, 2000.
2. K.C. John, Engineering Graphics, 2/e, PHI, 2013.